

509j CTE ALIGNMENT PROJECT

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CVHS Engineering & Applied Technology

College Now Eligibility Analysis

Six Courses • LBCC Articulation Pathways • Action
Recommendations

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About this document: This analysis was prepared as part of the 509j CTE Alignment process. Course content sourced directly from Mr. Kirsch's 2025-2026 class websites. LBCC program data from linnbenton.edu current catalog. College Now eligibility requires formal review by LBCC faculty and satisfies ODE CTE articulation requirements.

Executive Summary

What we found, what it means, what to do next.

Adam Kirsch teaches six distinct CTE-designated courses at Crescent Valley High School spanning applied engineering, woodworking, carpentry, and entrepreneurship. None currently carry College Now (dual credit) articulation with LBCC, despite significant content overlap with LBCC's Engineering Technology and CADD programs. This document evaluates each course against LBCC's current catalog and ranks articulation opportunity by strength of content match.

Course	Oregon Skill Set / Standards	Best LBCC Match	Match Strength	Priority
IDEA	NGSS HS-ETS; AZ CTE Eng. Sciences; CCSS Math	ENGR 248 — Engineering Graphics: Mechanical (SolidWorks, orthographic, CAD)	STRONG	#1 — Pursue First
DREAM	NGSS HS-ETS; AZ CTE Eng. Sciences; CCSS Math	ENGR 102 — Design Thinking & Problem Solving + BA 260 Entrepreneurship (partial)	MODERATE	#2 — Pursue with ENGR dept
Carpentry 1–3	Oregon Skill Set: Construction Cluster, Carpentry Focus (COPE10)	No direct LBCC match. CTE Construction apprenticeship pathway possible.	WEAK	#5 — Apprenticeship route
Woodworking 1	Oregon Skill Set: Manufacturing Cluster; AZ CTE Eng. Sciences	No dedicated LBCC woodworking program. MA3.396 Manufacturing Processes I (partial).	WEAK	#6 — Explore OSU Wood Sci.
Woodworking 2	OR Skill Set: Manufacturing (MNPI10) + Carpentry (COPE10); NGSS HS-ETS	EG4.412 — Inventor 3D Design (CAD component) MA3.420 CNC Operations (CNC component)	MODERATE	#3 — CAD component only
Woodworking 3 & 4	Oregon Skill Set: Manufacturing Cluster; Advanced CTE capstone	MA3.412 — Mastercam CAD/CAM MA3.420 — CNC Vertical Machining Ops	WEAK	#4 — CNC component only

LBCC does not currently offer a dedicated woodworking or carpentry certificate program. The strongest articulation opportunity is IDEA → ENGR 248. OSU's Wood Science and Engineering program is a longer-term pathway worth exploring for the woodworking sequence.

Key Findings

- **IDEA is the highest-priority course** for College Now pursuit — it already uses SolidWorks (the same software as ENGR 248), covers NGSS engineering design standards, earns students 1.5 HS credits, and has documented standards alignment.
- **DREAM is the strongest entrepreneurship match** but splits between engineering and business content. A split articulation (ENGR 102 + BA 260) may be most accurate.
- **The woodworking sequence has no direct LBCC path** — LBCC's programs focus on metal machining (Machine Tool Technology), not fine woodworking or furniture. The CNC and CAD sub-components of each course do have partial matches.
- **Carpentry aligns with Oregon's Construction Cluster skill set** (COPE10) but LBCC does not offer a stand-alone carpentry certificate — only apprenticeship-adjacent programs.
- **Kirsch carries 6 distinct courses** with no current College Now articulations. Adding even one (IDEA) would be a measurable win for the CTE alignment goals.

Course-by-Course Analysis

Each course evaluated against LBCC's current catalog for College Now eligibility.

COURSE 1 IDEA — Introduction to Design using Engineering and Art

Credits	Level	Prerequisites
1.5 HS credits (1.0 Science + 0.5 Applied Tech)	9th–10th grade entry point to engineering sequence	None listed — open enrollment gateway course

IDEA is a foundational engineering course earning students 1.5 high school credits (1.0 Science + 0.5 Applied Technology) — unusual for a single HS class and a strong signal of college-level rigor. Students work with industry-standard CAD software (SolidWorks), fabrication equipment (3D printers, laser engravers, CNC routers, UV printer, waterjet cutters, vinyl cutters), and cover physics, chemistry, statistics, and trigonometry in an engineering design context. Students routinely pass the CSWA (Certified SolidWorks Associate) exam.

Key Skills

Engineering design process, problem-solving, teamwork, technical communication, orthographic drawing, dimensioning, CAD, physics concepts (energy, waves, circuits), basic chemistry (stoichiometry), statistics, trigonometry, geometry

Software / Tools

SolidWorks 2019 (students pass CSWA exam), Adobe Illustrator, Mastercam, Z-Suite, Logger Pro (motion analysis), Inkscape

Standards

NGSS: HS-ETS1-1 through 4, HS-PS1, HS-PS2, HS-PS3, HS-PS4, HS-ESS3-4; CCSS Math: HSN.Q, HSA.SSE, HSA.CED, HSF.LE, HSS.ID, HSS.IC; AZ CTE Career Prep Standards: Engineering Sciences (all); CCSS Literacy in Science & Technical Subjects 9/10 (all)

Major Projects

Turbine design and analysis report; prototype fabrication using 3D printer, laser engraver, CNC router; physics motion analysis; multiple design challenges

STRONG MATCH ENGR 248 — Engineering Graphics: Mechanical (3 credits, LBCC Engineering program)

IDEA uses SolidWorks as its primary CAD tool; ENGR 248 specifies 'solid modelling/CAD software' — no brand requirement. SolidWorks and Inventor (LBCC's platform) are both parametric solid modelers; skills transfer directly. Both courses cover orthographic projection, dimensioning, and engineering graphics applied to real design problems. IDEA's documented NGSS HS-ETS standards directly parallel ENGR 248's engineering design content. The 1.5-credit HS structure demonstrates rigor above a typical elective. Kirsch holds a PE license (the national bar for engineering competency), fully satisfying LBCC instructor qualification requirements. Secondary option: ENGR 102 — Design Thinking & Problem Solving if ENGR 248 is too narrow.

To pursue articulation:

- Contact LBCC High School Partnerships (highschool@linnbenton.edu) to request ENGR 248 faculty review of IDEA syllabus
- Prepare syllabus document mapping IDEA content to ENGR 248 course outcomes
- Confirm Kirsch's credentials package: PE license (Oregon), engineering degree and industry experience — Kirsch meets LBCC instructor qualification requirements

- Submit formal College Now application through ODE's CTE articulation process
- Consider requesting dual articulation: ENGR 248 (CAD/design) + ENGR 111 (orientation)

COURSE 2 DREAM — Designing, Researching & Engineering Articles for Market

Credits	Level	Prerequisites
1.0 HS credit (Applied Technology / CTE)	11th–12th grade capstone; requires IDEA or equivalent	IDEA (or equivalent engineering coursework)

DREAM is the capstone of Kirsch's engineering sequence — students who have completed IDEA and other engineering courses apply all skills to design, prototype, and market real products using the full suite of school fabrication equipment. The course integrates engineering design, project management, community partnerships, and entrepreneurship. Products created in class are sold to support further course projects. Students also serve as mentors for younger students during 8th-grade engineering events.

Key Skills	Product design and optimization, engineering stress analysis, project management, marketing and product development, technical drawings and dimensioning, teamwork and leadership, employability skills, soldering and basic electronics, community engagement, mentorship of younger students
Software / Tools	SolidWorks 2019, Adobe Illustrator, Mastercam, Sheetcam, Inkscape; Full equipment: 3D printers, laser engraver, vinyl cutter, CNC routers, UV printer, Shaper Origin
Standards	NGSS: HS-ETS1-1 through 4, HS-PS1, HS-PS2, HS-PS3, HS-PS4, HS-ESS3-4; CCSS Math: full HSN, HSA, HSF, HSS series; AZ CTE Career Prep Standards: Engineering Sciences (all); CCSS Literacy in Science & Technical Subjects 9/10 (all, including WHST.3)
Major Projects	Community-based engineering challenges; product prototyping and sale; scavenger hunt activity design for 8th-grade visitors; project timelines and management documentation; internships with community business partners

MODERATE MATCH ENGR 102 — Design Thinking & Problem Solving + BA 260 — Entrepreneurship (LBCC Engineering + Business programs — split articulation)

DREAM's engineering component (design process, CAD, fabrication) aligns with ENGR 102, while its product-to-market and project management content aligns with BA 260 (Entrepreneurship & Small Business). A single LBCC course that covers both the technical and entrepreneurial content does not currently exist, making a split articulation the most honest representation. ENGR 102 alone would undervalue the course; BA 260 alone would misrepresent it as business-only. Best strategy: pursue ENGR 102 first (as it builds on the IDEA → ENGR 248 relationship), then explore BA 260 as an additional articulation in a future cycle.

To pursue articulation:

- Build on the IDEA → ENGR 248 relationship first; establish Kirsch as a College Now instructor before adding DREAM
- Request LBCC Engineering faculty review of DREAM for ENGR 102 equivalency
- Contact LBCC Business department about BA 260 review for the entrepreneurship component
- Document the product-to-market project cycle with photos and sales records — this is compelling evidence of college-level entrepreneurship rigor
- Consider pursuing ENGR 102 in year 2 after IDEA is established

COURSE 3 Woodworking 2

Credits	Level	Prerequisites
1.0 HS credit (CTE — Manufacturing/Construction)	Intermediate; requires Woodworking 1 or equivalent	Woodworking 1 or demonstrated shop skills

Woodworking 2 is an intermediate hands-on course explicitly aligned to Oregon's CTE Skill Sets in both the Manufacturing Cluster (Wood Products) and Construction Cluster (Carpentry). Students work with raw lumber, advanced joinery, CAD-designed projects, CNC machining (V-Carve Pro), and wood finishing. The course emphasizes project planning, material estimation, technical drawing, and professional presentation at the Mid Willamette Valley Woodworkers Guild show.

Key Skills	Wood identification and properties, material cost estimation, advanced joinery (mortise/tenon, dovetails, inlay), multi-view technical drawing, wood finishing, project planning (POP template), safe tool/machine operation, wood science basics
Software / Tools	SolidWorks 2019, V-Carve Pro (CNC), Inkscape, Adobe Illustrator
Standards	Oregon Skill Set: Manufacturing Cluster, Wood Products (MNPI10.01–10.03); Oregon Skill Set: Construction Cluster, Carpentry (COPE10.01–10.02); NGSS: HS-PS2-6, HS-ETS1-2, HS-ETS1-3; AZ CTE Career Prep: Engineering Sciences 1.0–6.0
Major Projects	Complex inlay project (CNC + V-Carve); two personal projects (100+ pts each); submission to Mid Willamette Valley Woodworkers Guild show (public library, April)

MODERATE MATCH EG4.412 — Inventor 3D Design (CAD component only) (LBCC CADD program) + MA3.420 — CNC Operations (CNC component)

LBCC has no dedicated woodworking or wood technology program, which is the core challenge. However, two sub-components of Woodworking 2 have meaningful LBCC equivalents: (1) The CAD/technical drawing work aligns with EG4.412 (Inventor 3D Design) or ENGR 248 — both use parametric solid modeling and dimensioning standards. (2) The V-Carve/CNC machining work aligns with MA3.420 (CNC Vertical Machining Operations). A full-course articulation is not viable — the woodworking-specific content has no LBCC equivalent. A partial or component articulation is the realistic near-term goal.

To pursue articulation:

- Identify which portion of Woodworking 2 hours are spent on CAD (SolidWorks/V-Carve) vs. hand/power tool work — this determines if a CAD-only articulation is defensible
- Explore whether OSU's Wood Science and Engineering (BS program) has community college pathways or articulation agreements that CVHS could connect to
- Contact Oregon Dept. of Education CTE office about the Wood Products skill set — they may know of other CC partners with woodworking programs
- In the near term, document Oregon Skill Set alignment (MNPI10/COPE10) formally — this is required for any future articulation and good CTE practice regardless

COURSE 4 Woodworking 3 & 4

Credits	Level	Prerequisites
1.0 HS credit (CTE — Advanced Manufacturing/Construction)	Advanced capstone; requires Woodworking 1 & 2	Woodworking 1 and 2; safety certification

Woodworking 3 & 4 is the advanced/capstone level of the woodworking sequence. Students complete three 300-point projects demonstrating mastery of joinery, design, and professional-quality finishing. Students also complete the Forest Service Trail Signs project (real commissioned work for the National Forest) using V-Carve CNC routing, and submit work to the Mid Willamette Valley Woodworkers Guild public show. Safety certification is required before any equipment use.

Key Skills	Advanced joinery and hand tool mastery, professional finishing, CNC routing and toolpath generation (V-Carve Pro), G-code file preparation, project management (POP/timeline documentation), safety certification across all major equipment, professional presentation for guild show
Software / Tools	V-Carve Pro, SolidWorks, Inkscape; Full shop: band saw, table saw, router, drill press, CNC router, lathe
Standards	Oregon Skill Set: Manufacturing Cluster (advanced); CVHS Dimensioning Standards; POP/portfolio documentation standards; Real-world commission work (Forest Service Trail Signs)
Major Projects	Forest Service Trail Signs (commissioned real-world project); three 300-point personal projects; Mid Willamette Valley Woodworkers Guild show submission (2 pieces, public library, April 23-26)

WEAK MATCH MA3.412 — Mastercam CAD/CAM Programming (CNC component) (LBCC Machine Tool Technology program)

The same LBCC gap applies here as with Woodworking 2: no woodworking program exists. The CNC routing work in this course uses V-Carve (wood-specific), not Mastercam (metal-focused) — so even the CNC match is imperfect. MA3.412/MA3.420 target metal CNC machining, not wood CNC routing. This course's strongest value proposition is the real-world commissioned work (Forest Service Trail Signs) and guild show participation — hallmarks of a high-quality CTE capstone — but without a LBCC woodworking program, formal articulation is not currently viable.

To pursue articulation:

- Document the Forest Service Trail Signs as an industry partnership — this is significant evidence for any future articulation or grant application
- Long-term: explore Chemeketa Community College (Salem) or Lane Community College (Eugene) — both may have cabinet/woodworking programs that could articulate
- Consider Oregon apprenticeship pathway: Oregon Building Industry Association / Carpenters Union apprenticeship programs accept HS credit through ODE
- For now: ensure Oregon Skill Set documentation is current and complete — required for CTE designation regardless of College Now status

COURSE 5 Carpentry 1–3

Credits	Level	Prerequisites
1.0 HS credit per level (CTE — Construction)	Levels 1, 2, and 3 — multi-year sequence	None for Carpentry 1; subsequent levels build on prior

The Carpentry sequence centers on real construction work — students build actual micro-shelters and tiny homes donated to the community (Unity Shelter). Carpentry 1 focuses on structural work: roofing (drip edge, underlayment, shingles), Hardi panel soffits, window and door installation. Carpentry 2 & 3 add CNC design work with V-Carve Pro and phone holder fabrication. Students maintain documentation portfolios via Google Slides and mentor younger students in pre-engineering activities. The program also connects to OSU's Wood Science and Forestry programs.

Key Skills	Structural carpentry (roofing, siding, windows, doors, framing), tool safety and maintenance, shop cleanliness and workplace readiness, CNC/V-Carve design, technical drawing (SolidWorks), photography documentation, project portfolio, mentorship and teaching younger students
Software / Tools	V-Carve Pro, SolidWorks 2019, Inkscape, Adobe Illustrator; Physical: framing tools, roofing equipment, power saws, drill
Standards	Oregon Skill Set: Construction Cluster, Carpentry Focus (COPE10.01–10.02); Pre-Engineering Program Syllabus standards; Real construction documentation requirements
Major Projects	Micro-shelter/tiny home construction (Shelter #9 completion, Shelter #10 build); Roofing installation; CNC phone holder design (Carpentry 2&3); Pre-engineering mentorship for 8th-grade Open House

WEAK MATCH No direct match — LBCC lacks a carpentry certificate program (Oregon Building Trades Apprenticeship pathway — alternative route)

LBCC does not offer a carpentry or residential construction certificate. Their construction-adjacent offerings are within the Engineering program (Construction Engineering emphasis — ENGR 245 Engineering Graphics: Civil) and apprenticeship support programs. The Carpentry course's real construction work (tiny homes for Unity Shelter) is exceptional CTE practice and community partnership — but without a LBCC carpentry equivalent, College Now articulation is not currently possible. The most viable alternative is Oregon's registered apprenticeship pathway, where documented HS construction hours can count toward Carpenters Union apprenticeship.

To pursue articulation:

- Contact Oregon BOLI (Bureau of Labor & Industries) Apprenticeship Division about pre-apprenticeship credit for documented carpentry hours
- Reach out to the Mid-Willamette Valley Carpenters Local Union about partnership
- Explore ENGR 245 (Engineering Graphics: Civil) articulation for any structural drawing/design component if Carpentry courses include civil drawing work
- Document Unity Shelter partnership formally — this is a powerful community partnership that strengthens any grant or program review
- Check whether Chemeketa CC's Construction program offers College Now agreements with other districts — if so, pursue a similar agreement for 509j

COURSE 6 Woodworking 1

Credits	Level	Prerequisites
1.0 HS credit (CTE — Manufacturing/Construction)	Entry-level; no prerequisites	None — open enrollment

Woodworking 1 is the entry-level course in Kirsch's woodworking sequence. Students learn foundational woodworking — hand tools, power tools, wood selection, and basic project construction — alongside introductory CNC skills using V-Carve Pro. The Forest Service Trail Signs project begins here, with students creating two signs using V-Carve software for CNC engraving and submitting work to the Mid Willamette Valley Woodworkers Guild show. Community connections include Oregon State University Wood Science and Engineering and local sawyer/supplier partnerships.

Key Skills	Hand tool operation and safety, power tool operation, wood selection and properties, project planning (POP template), V-Carve Pro for CNC design (toolpaths, G-code), basic Solidworks CAD, portfolio documentation
Software / Tools	V-Carve Pro, SolidWorks 2019, Inkscape, Adobe Illustrator
Standards	Oregon Skill Set: Manufacturing Cluster; AZ CTE Eng. Sciences (partial); CVHS safety and dimensioning standards
Major Projects	Forest Service Trail Signs (V-Carve CNC engraving, due for guild show); foundational woodworking projects (100-point level); Mid Willamette Valley Woodworkers Guild show submission

WEAK MATCH No direct match — MA3.396 Manufacturing Processes I is closest (partial) (LBCC

Machine Tool Technology — wood-specific content not covered)

Woodworking 1 is foundational and introductory — even if LBCC had a woodworking program, a 100-level entry course typically does not qualify for college-level dual credit. The CNC and CAD elements are taught at a basic level in this course (covered more deeply in Woodworking 2 and 3&4). MA3.396 (Manufacturing Processes I) covers metalworking processes, not wood. This course is best understood as building toward the Woodworking 2/3&4 sequence rather than a standalone College Now target.

To pursue articulation:

- Focus College Now energy on Woodworking 2 and 3&4 rather than Woodworking 1 — the content is more advanced and defensible
- Ensure Woodworking 1 is formally documented as a CTE course under Oregon Skill Sets so it counts toward pathway completion metrics
- Track student progression through the full sequence (1 → 2 → 3&4) — longitudinal data strengthens any future articulation case

Action Plan & Next Steps

Prioritized recommendations for pursuing College Now articulation.

Based on content strength and LBCC program availability, here is a phased approach to building College Now articulations for Kirsch's courses over 2–3 years.

Year 1 — Immediate (2025-26)

- Submit IDEA → ENGR 248 articulation request to LBCC (highschool@linnbenton.edu). Prepare syllabus mapping to LBCC course outcomes.
- Compile Kirsch's credentials package: PE license (Oregon), engineering degree and industry experience, CSWA student outcomes — all included in Kirsch_Credentials.docx.
- Formally document Oregon Skill Set alignment for ALL six courses — required for CTE designation and a prerequisite for any articulation.
- Document the Unity Shelter tiny home partnership and Forest Service Trail Signs as industry/community partnerships (required for Perkins V reporting).

Year 2 — Build (2026-27)

- After IDEA is established, pursue DREAM → ENGR 102 with LBCC Engineering faculty.
- Contact Oregon BOLI Apprenticeship Division about pre-apprenticeship credit for Carpentry hours (documented construction hours toward Carpenters Union apprenticeship).
- Research Chemeketa CC (Salem) and Lane CC (Eugene) for woodworking/cabinetry programs — if either has College Now agreements with other Oregon districts, 509j can follow.
- Explore OSU Wood Science articulation pathway for the Woodworking sequence capstone.

Year 3 — Expand (2027-28)

- If LBCC adds a Construction or Wood Technology program, pursue Carpentry and Woodworking articulations formally.
- Pursue BA 260 (Entrepreneurship) articulation for the DREAM business component.
- Evaluate whether Woodworking 2's CAD/CNC content can articulate as a component credit alongside a full woodworking articulation from another CC.
- Review all articulations annually per ODE CTE program review requirements.

Key Contacts for Articulation Pursuit

Contact	Role	Relevance
highschool@linnbenton.edu	LBCC High School Partnerships	First contact for all College Now inquiries — IDEA, DREAM
LBCC Engineering Dept.	ENGR 248 / ENGR 102 faculty	Syllabus review and equivalency determination for IDEA/DREAM
LBCC CADD Program	EG4.412 faculty	CAD component review for Woodworking 2 (longer-term)
Oregon BOLI Apprenticeship (503) 378-3272	Pre-Apprenticeship Programs	Carpentry construction hours toward Carpenters Union apprenticeship

ODE CTE Office
ode.oregon.gov/cte

Oregon CTE Standards &
Articulation

Oregon Skill Set documentation, Perkins V compliance,
articulation guidance

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